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QUANTITATIVE RISK & BEHAVIOURAL AI · CASE STUDY

Lolo's Ledger: Multi-Asset Risk Engine & Tilt Detector

BI Scientist • AI/ML Engineer • Python Developer

Multi-asset trading intelligence platform enforcing dynamic risk sizing, revenge-trading Tilt Index detection, and personal-edge attribution.

BEHAVIOURAL PSYCHOLOGY

Why Traders Lose: The Revenge-Trade Loop

Trading failures stem not from bad technical indicators, but from uncontrolled position sizing and emotional tilt after a loss.

Revenge Sizing

- Trader suffers a loss and re-enters the market within 15 minutes.
- Doubles position size to 'win back' lost capital.
- Results in rapid account blowouts and cascading drawdowns.

No Edge Validation

- Traders jump between setups without statistical verification.
- Confuse a lucky winning streak with a genuine mathematical edge.
- Overfit strategies to tiny sample sizes (N < 30 trades).

Multi-Asset Complexity

- Pip calculations in FX differ completely from point values in Indices or Crypto.
- Manual lot sizing leads to catastrophic calculation errors.
- Requires hard mathematical position-sizing guardrails.

QUANTITATIVE ARCHITECTURE

Multi-Asset Risk Guardrails & Behavioural Tilt Detector

Deterministic risk engine backed by strict ethical guardrails: zero buy/sell signals generated.

Multi-Asset Risk Guard

- Strict 0.25%–2.0% risk cap per trade.
- Automated lot sizing for FX, Crypto, Indices, and Equities.
- 4% daily drawdown circuit breaker.

Behavioural Tilt Index

- Monitors post-loss cooldown periods.
- Flags stop-loss widening and overtrading.
- Quantifies cumulative 'tilt tax' in R-multiples.

Edge Attribution (N ≥ 100)

- Strict sample size gate: refuses fit if N < 100.
- Calibrated logistic model attributing edge factors.
- 100% focused on trader process, not price prediction.

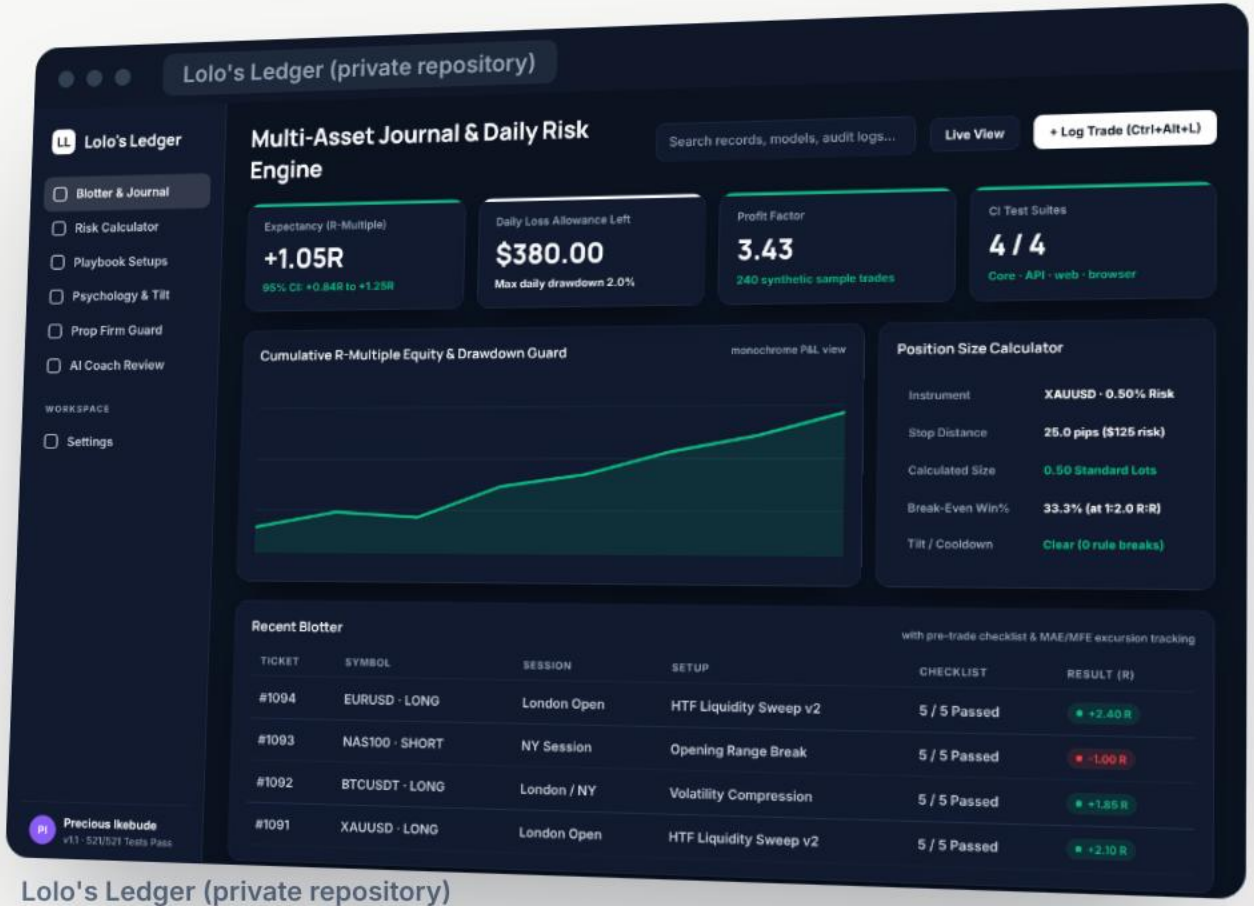
EXECUTION PLATFORM

Lolo's Ledger Web App

Trading journal and quant engine, tested end to end in CI (core, API, web and browser suites).

- **Cross-Asset Position Calculator:** Select any pair (EURUSD, XAUUSD, US100, BTCUSDT) to get instant, exact lot sizing based on account balance and stop loss.
- **Expectancy Engine:** Computes E[R] with a confidence interval, profit factor, Sharpe/Sortino, MAE/MFE and session win rates.
- **Tilt Detection:** Flags revenge re-entries, stop-loss widening and overtrading, and shows what they cost in R.

Python Quant | TypeScript | React + NestJS | CI-tested



Lolo's Ledger (private repository)

ENGINE CHECKS

Risk Engine on a $10,000 Account

Worked examples from reports/metrics.json. The journal analytics run on a synthetic sample journal; they show the engine works, not trading results.

EURUSD, 1% risk

0.50 lots

20-pip stop, $100 at risk

XAUUSD, 1% risk

0.20 lots

500-tick stop, $100 at risk

Edge Model Gate

N ≥ 100

Refuses to fit on fewer trades

Risk Cap

0.25–2%

Per trade; 4% daily loss limit

Tilt Detection

- Flags re-entries within 15 minutes of a loss.
- Flags widened stop-losses and bursts of overtrading.
- Reports the cost of flagged trades in R for the trader's own journal.

Ethical AI Guardrails

- The AI coach never gives buy/sell signals or price predictions.
- It explains the trader's own data: educational, not financial advice.
- Playwright end-to-end tests at phone and desktop widths.

PERSONAL PRODUCT

Discipline engineered into software.

Lolo's Ledger puts position sizing and behavioural guardrails in front of the trader, and never tells them what to buy or sell.

Author: Precious Ikebude
Code: private repository; walkthrough on request